ORGANIC CHEMISTRY

ROADMAP PROBLEM # 01 [SOLUTION]

A : Sucrose (+)

A_1 : Glucose (+)

A_2 : Fructose (–)

Black particles : Carbon (B)

C : CaO

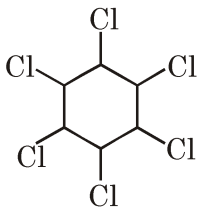
D : CaC_2

E : $HC \equiv CH$

F :

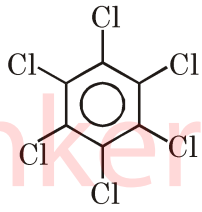


G :



BHC
Benzene
hexachloride

H :

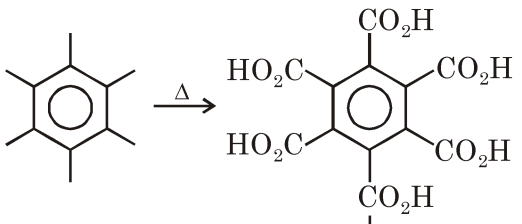


Hexachloro
benzene

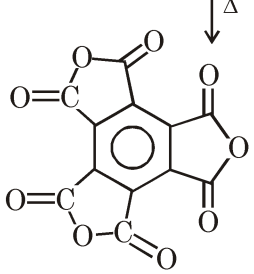
I : $\overset{+}{N}a\overset{\ominus}{C}\equiv\overset{\ominus}{C}\overset{+}{N}a$

J : $H_3C - C \equiv C - CH_3$

K



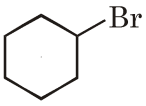
L



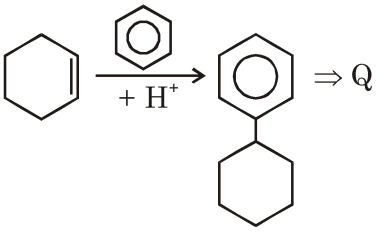
M :



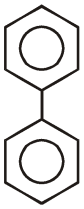
N :



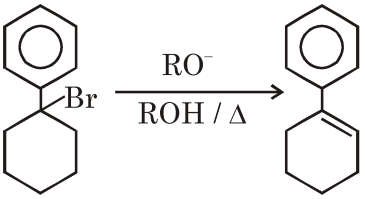
O :



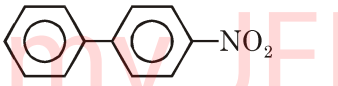
P :



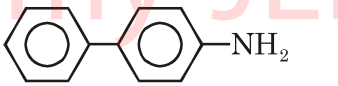
R :



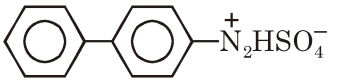
S :



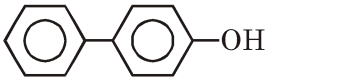
T :



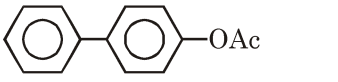
U :



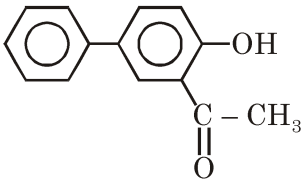
V :



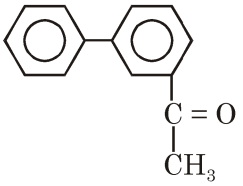
W :



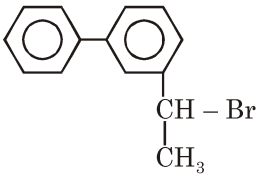
X :



Y :



Z :



Rankers Academy JEE

ORGANIC CHEMISTRY

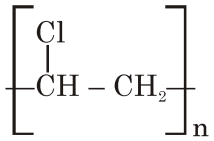
ROADMAP PROBLEM # 02 [SOLUTION]

A : D-Mannose

B : H–CHO Formaldehyde

C : H–COOH

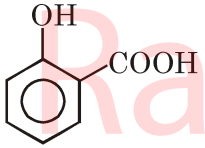
D : $CH_3 - OH$ Methyl alcohol

E :  Vinyl chloride

F : H–COOMe Methyl formate

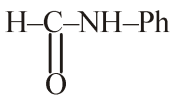
G : Ph – CHO Benzaldehyde

H : MeOMgBr

I :  Salicylic acid

J : Me – CHO Acetaldehyde

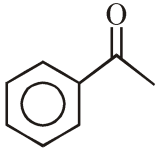
K : Ph–CH=N–OH

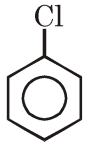
L : 

M : Ph–C≡N Benzonitrile

N :  n-hexane

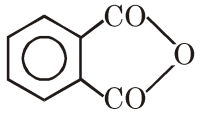
O :  Benzene

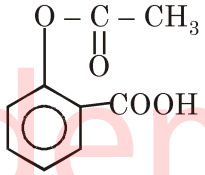
P :  Acetophenone

Q :  Chlorobenzene

R : Ph – MgCl

S : Ph – OH Phenol / Carbolic acid

T :  Phthalic anhydride

U :  Aspirin

V : Ph – COOH Benzoic acid

W : HC ≡ CH Acetylene

X : D-Glucose

Y : D-Fructose

Z : Ph – CH_3 Toluene

Rankers Academy JEE

ORGANIC CHEMISTRY

ROADMAP PROBLEM # 03 [SOLUTION]

P : CH_3CH_2OH

Q : CH_3CHO

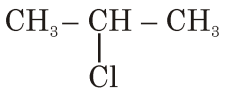
R : $CH_3 - CH = CH - CHO$

S : $CH_3CH_2CH_2CH_2OH$

T : $CH_3CH_2CH_2COOH$

U : $CH_3CH_2CH_3$

V :



W : $CH_3 - CH = CH_2$

X : $HCHO$

Y : CH_3CHO

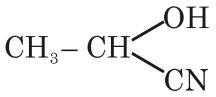
Z : CH_3OH

A : CH_3Cl

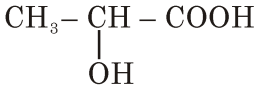
B : CH_3CH_3

C : CH_3CH_2Br

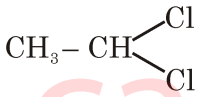
D :



E :



F :



G : $CH_2=CH-Cl$

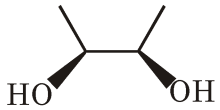
H : $Na^+\bar{C}\equiv\bar{C}Na^+$

I : $CH_3-C\equiv C-CH_3$

J :



K :



L : CH_3-CHO

M : CH_3-COOH

N : CH_3COOEt

O :



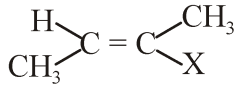
Rankers Academy JEE

ORGANIC CHEMISTRY

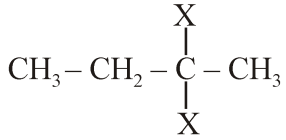
ROADMAP PROBLEM # 04 [SOLUTION]

A : $CH_3–C\equiv C–CH_3$ (But-2-yne)

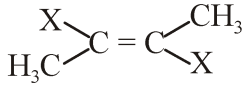
B :



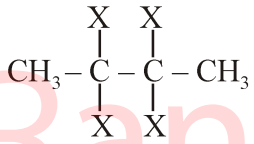
C :



D :

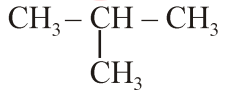


E :

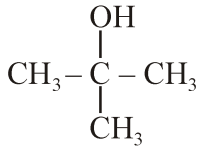


F : $CH_3–CH_2–CH_2–CH_3$

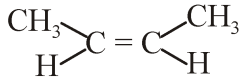
G :



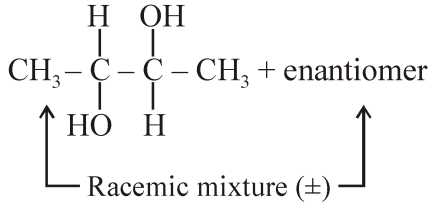
H :



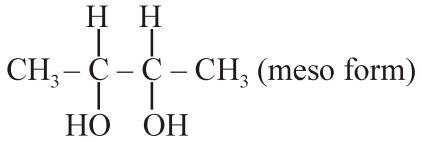
I :



J :



K :



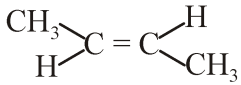
L : $CH_3–CH_2OH$

M : $CH_3–CHO$

N : $CH_3–CH=CH–CHO$

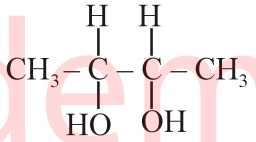
O : $CH_3–CH_2–CH_2–CH_2–OH$

P :

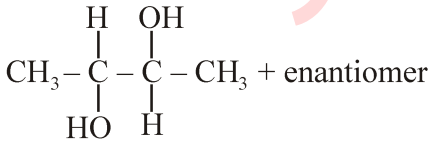


Q : $CH_3–COOH$

S :



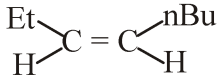
R :



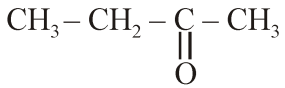
T : $CH_3–CH_2–C\equiv CH$

V : $CH_3–CH_2–C\equiv C^-Ag^+\downarrow$

U :



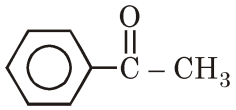
W :

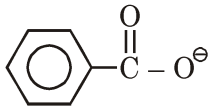


Rankers Academy JEE

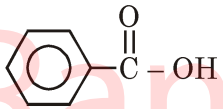
ORGANIC CHEMISTRY

ROADMAP PROBLEM # 05 [SOLUTION]

A : 

B : 

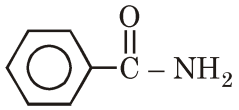
C : CHI_3

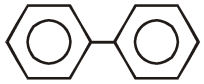
D : 

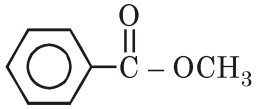
E : $CH \equiv CH$

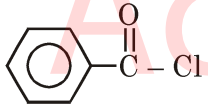
F : 

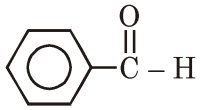
G : 

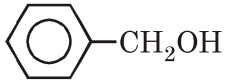
H : 

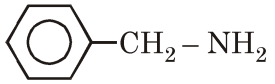
I : 

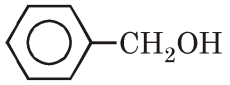
J : 

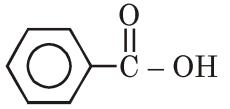
K : 

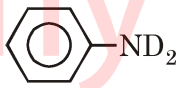
L : 

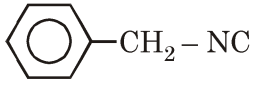
M : 

N : 

O : 

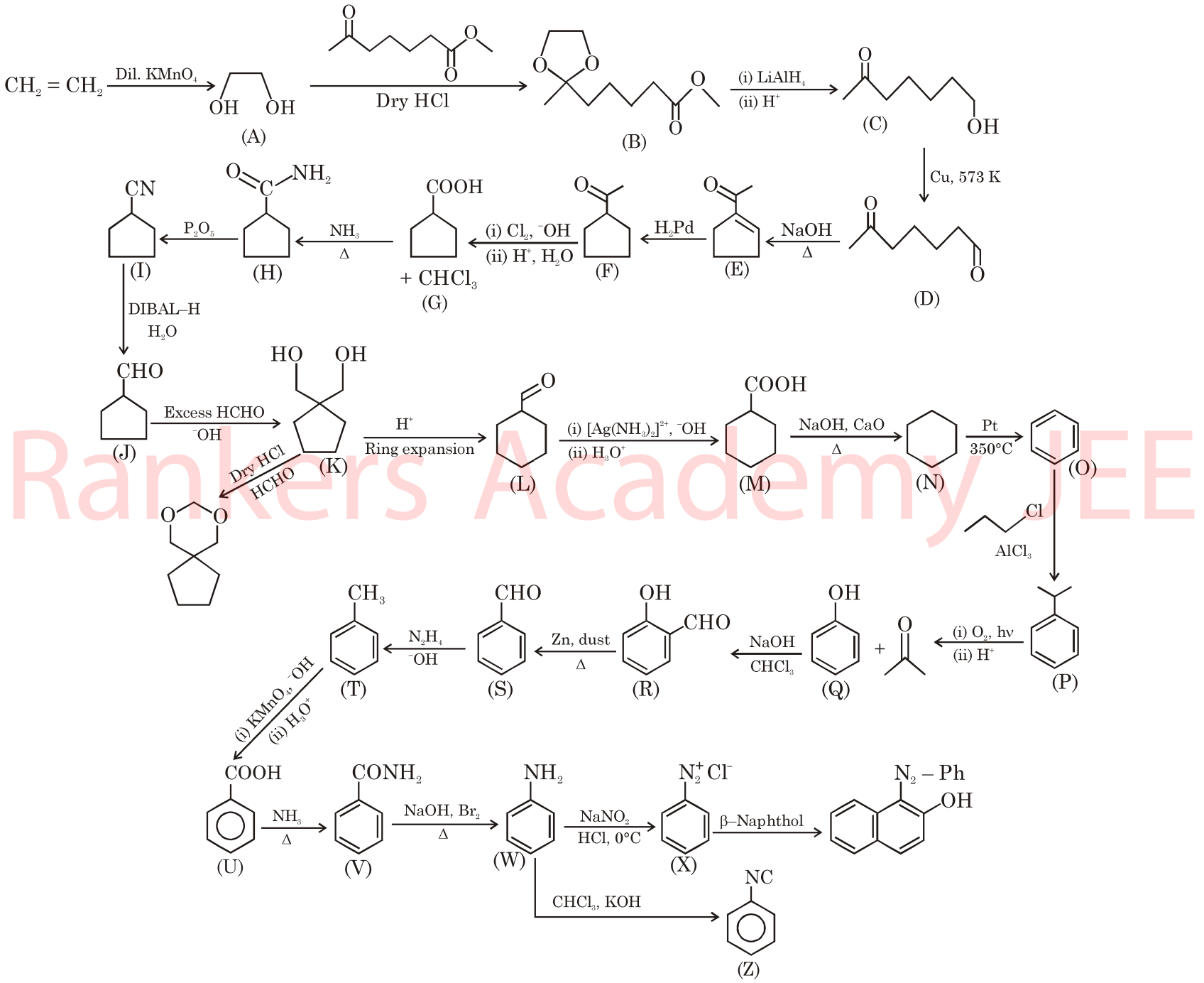
P : 

Q : 

R : 

Rankers Academy JEE

SUBJECTIVE ROAD MAP PROBLEM SOL_6

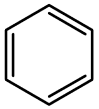


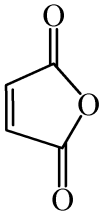
Rankers Academy JEE

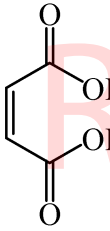
ORGANIC CHEMISTRY

ROADMAP PROBLEM # 07 [SOLUTION]

A. $CH \equiv CH$

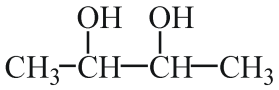
B. 

C. 

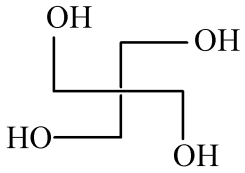
D. 

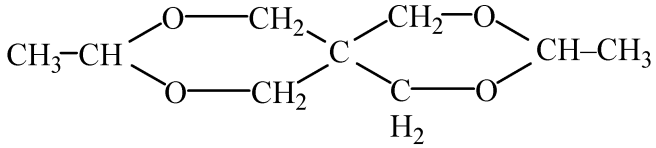
E. $CH \equiv CH$

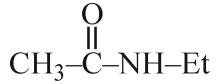
F. $CH_3–CHO$

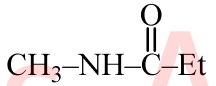
G. 

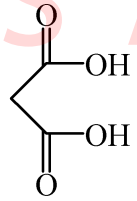
H. 

I 

J. 

K. 

L. 

M. 

N. $CH_3–COOH$

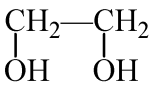
O. CH_4

P. $CH_3–Br$

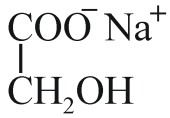
Q. $CH_3–CH_3$

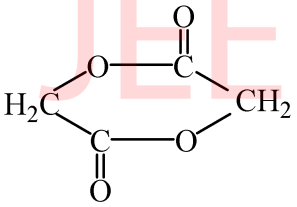
R. $CH_3–CH_2–Br$

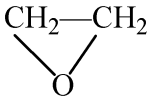
S $CH_2=CH_2$

T 

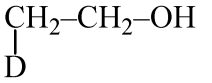
U 

V 

W 

X 

Y $Ph–CH_2–CH_2–OH$

Z 

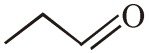
Rankers Academy JEE

Answer iupac name

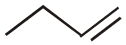
(C) But-2-ene-1,4-di, oicanhidride
(D) But-2-ene-1,4-dioic acid
(G) Butane-2,3-diol
(I) 2,2-dihydroxymethylpropan-1,3-diol
(K) N-ethyl ethanamide
(L) N-methyl propanamide
(M) Propan-1,3-dioic acid
(U) Ethan-1,2-dial
(X) 1,2-epoxy ethane
(Y) 2-Phenylethanol
(Z) 2-Deuteroethan-1-ol

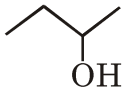
ORGANIC CHEMISTRY

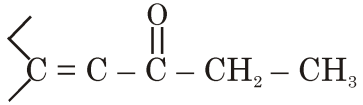
ROADMAP PROBLEM # 08 [SOLUTION]

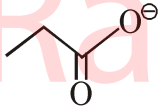
(A) 

(B) 

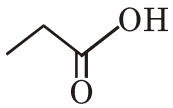
(C) 

(D) 

(E) 

(F) 

(G) CHI_3

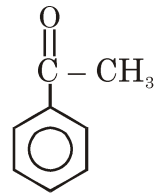
(H) 

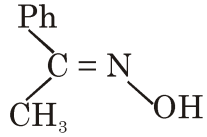
(I) 

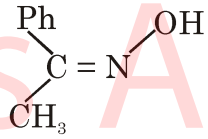
(J) 

(K) $CH \equiv CH$

(L) 

(M) 

(N) 

(O) 

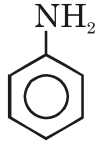
(P) 

(Q) $Ph - CO_2H$

(R) $CH_3 - NH_2$

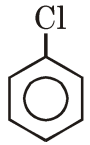
(S) $CH_3 - NC$

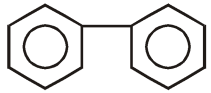
(T) 

(U) 

(V) $Ph - N_2^+Cl^\ominus$

(W) CH_4

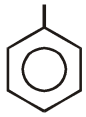
(X) 

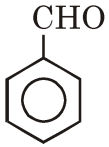
(Y) 

Rankers Academy JEE

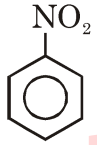
ORGANIC CHEMISTRY

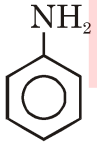
ROADMAP PROBLEM # 09 [SOLUTION]

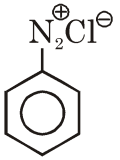
(A) 

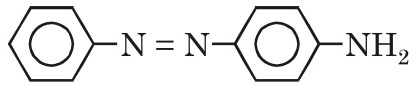
(B) 

(C) $Ph - CH = CH - COOH$

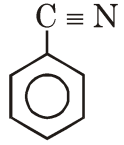
(D) 

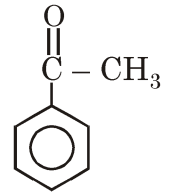
(E) 

(F) 

(G) 

(C–N Coupling)

(H) 

(I) 

(J) $CHCl_3$

(K) $H-C \equiv C-H$

(L) $CH_3 - \underset{\underset{O}{\|}}{C} - H$

(M) $CH_3 - CH = CH - \underset{\underset{O}{\|}}{C} - H$

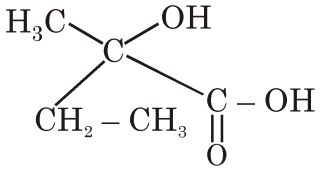
(N) $CH_3 - C \equiv C - CH_3$

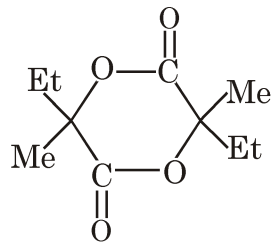
(O) 

(P) $CH_3 - \underset{OH}{CH} - \underset{OH}{CH} - CH_3$

(Q) $CH_3 - \underset{\underset{O}{\|}}{C} - CH_2 - CH_3$

(R) $CH_3 - \underset{\underset{C \equiv N}{|}}{\overset{\overset{OH}{|}}{C}} - CH_2 - CH_3$

(S) 

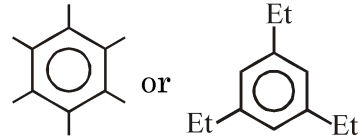
(T) 

(U) $CH_3 - CH_2 - \underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}} - CH_2 - OH$

(V) $CH_3 - \overset{\overset{O}{\|}}{C} - CH_2 - CH_3$

(W) $H - \overset{\overset{O}{\|}}{C} - H$

(X) $CH_3 - OH$

(Y) 

(Z) 

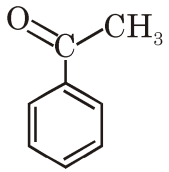
Rankers Academy JEE

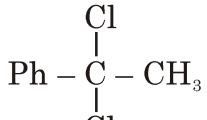
ORGANIC CHEMISTRY

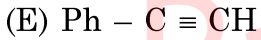
ROADMAP PROBLEM # 10 [SOLUTION]

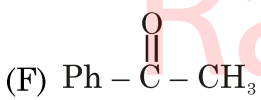
(A) C_2H_2

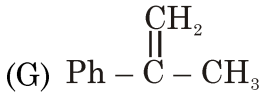
(B) 

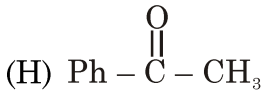
(C) 

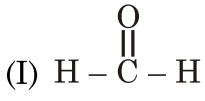
(D) 

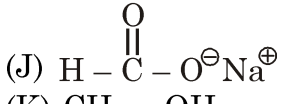
(E) 

(F) 

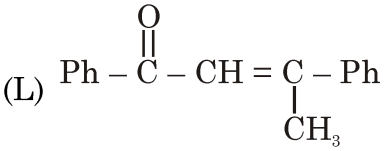
(G) 

(H) 

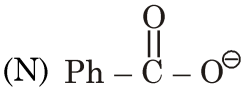
(I) 

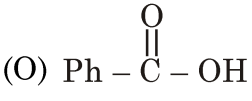
(J) 

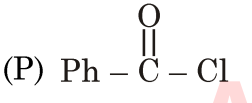
(K) $CH_3 - OH$

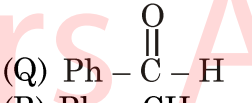
(L) 

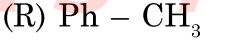
(M) CHI_3 (yellow ppt)

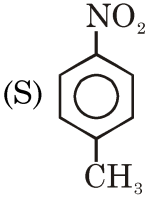
(N) 

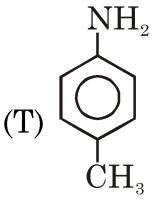
(O) 

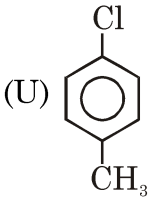
(P) 

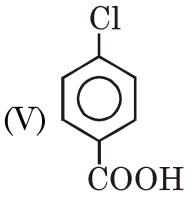
(Q) 

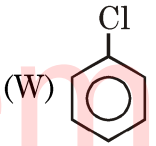
(R) 

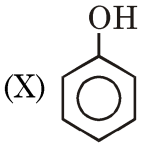
(S) 

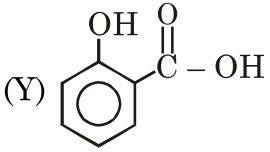
(T) 

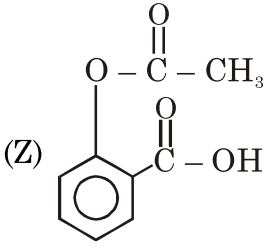
(U) 

(V) 

(W) 

(X) 

(Y) 

(Z) 

Rankers Academy JEE

ORGANIC CHEMISTRY

ROADMAP PROBLEM # 11 [SOLUTION]

A $CH_3 - CH_2 - COOH$

B $CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - COOH$ HVZ reaction

C $CH_3 - \underset{\displaystyle NH_2}{\underset{|}{CH}} - COOH$ Alanine

D $CH_3 - CH_2 - NH_2$ Schmidt degradation

E $Ph - SO_2 - NH - Et$

F $CH_3 - CH_2 - OH$ Diazotisation

G $CH_3 - CH_2 - COOEt$ Esterification

H $CH_3 - CH_2 - \overset{\displaystyle O}{\overset{\|}{C}} - \underset{\displaystyle CH_3}{\underset{|}{CH}} - COOEt$ Claisen condensation

I $CH_3 - CH_2 - \overset{\displaystyle O}{\overset{\|}{C}} - CH_2 - CH_3$

J $CH_3 - CH_2 - \overset{\displaystyle O}{\overset{\|}{C}} - Cl$

K $CH_3 - CH_2 - CH_2 - COOH$ Arndt Eistert reaction

L $CH_3 - CH_2 - Br$ Hundsdeicker reaction

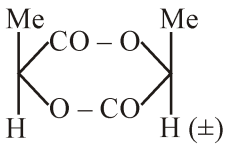
M $CH_3 - CH = O$

N $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CN$ Cyanohydrin formation

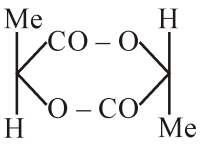
O $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - COOH$ Lactic acid

P $CH_3 - \overset{\displaystyle O}{\overset{\|}{C}} - COOH$ Pyruvic acid

Q_1, Q_2 and Q_3 is a mixture of three lactides.
Here it is heating effect of α-hydroxy acid is applied.



Q_1 / Q_2
optically active



Q_3 meso
due to COS

R CHI_3

Rankers Academy JEE

S  oxalic acid

Formation of R and S is Iodoform reaction

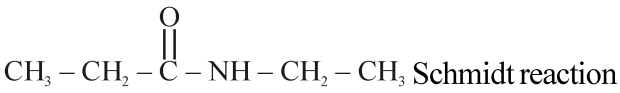
T HCOOH (Heating effect on oxalic acid)

U $HC \equiv CH$

V $H_2C = CH - COOH$ Acrylic acid

W $H_2C = CH - COOMe$ methyl acrylate

This on polymerisation gives polymethyl acrylate (PMA) substance used to make acrylic glasses.

X  Schmidt reaction

Y $CH_3 - CH_2 - NC$ Carbylamine reaction

Z $CH_3 - CH_2 - CH_2 - CH_3$ Kolbe's electrolysis.

Rankers Academy JEE

ORGANIC CHEMISTRY

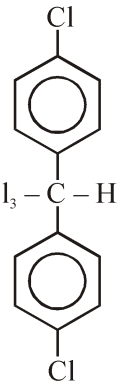
ROADMAP PROBLEM # 12 [SOLUTION]

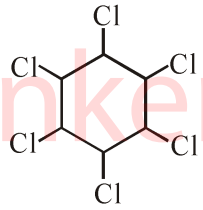
A Glucose

B n-Hexane

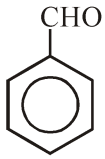
C Benzene

D 

E (D.D.T.) $CCl_3 - \underset{H}{\overset{|}{C}} -$  % of Cl ≈ 50%

F  (B.H.C.) total G.I. → 8

Rankers Academy JEE

G 

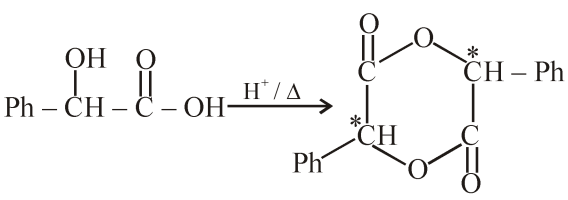
H $Ph - CH = CH - COOH$

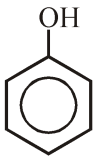
I $Ph - \underset{Br}{\overset{|}{C}}H - \underset{Br}{\overset{|}{C}}H - COOH$

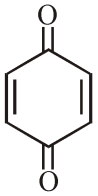
J $Ph - C \equiv C - H$

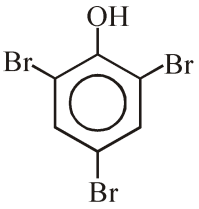
K $Ph - \overset{O}{\overset{\|}{C}} - CH_3$

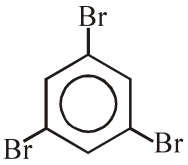
L $Ph - \overset{O}{\overset{\|}{C}} - \overset{O}{\overset{\|}{C}} - H$

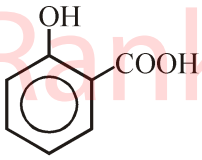
M  total product(s) = 3

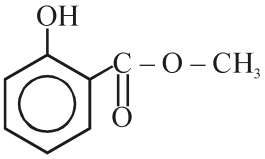
N 

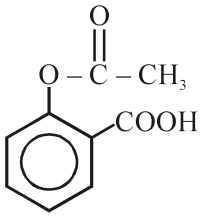
O  D.U. = 5

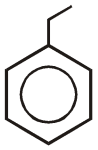
P 

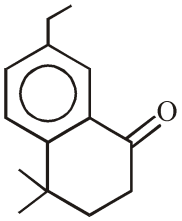
Q 

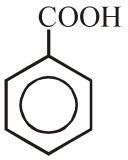
R 

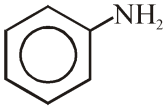
S 

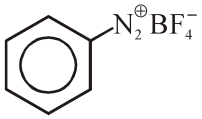
T 

U 

V 

W 

X 

Y 

Z 

Rankers Academy JEE

Extra question answer

1. If E.W.G. present on benzene ring, then ozonolysis of benzene is not performed.

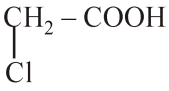
ORGANIC CHEMISTRY

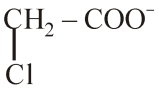
ROADMAP PROBLEM # 13 [SOLUTION]

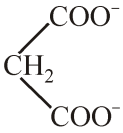
A CH_4

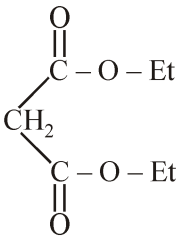
B CH_3Cl

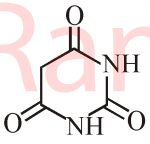
C $CH_3 - COOH$

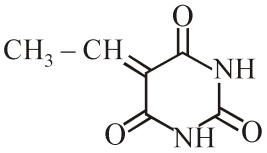
D 

E 

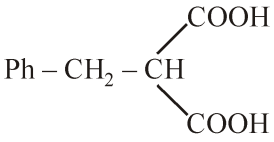
F 

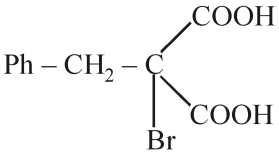
G 

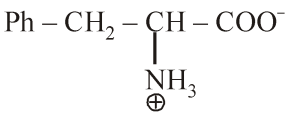
H 
(Barbituric acid) act as Tranquilizer

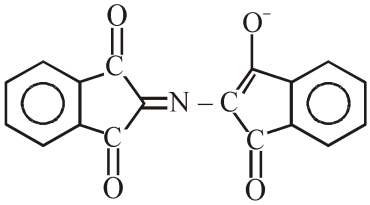
I 

%N $\Rightarrow$ 18.18

J 

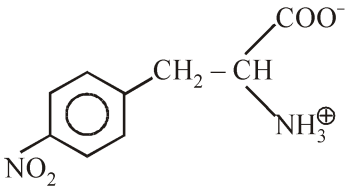
K 

L 

M 

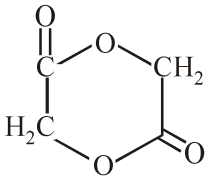
(Purple-colour ion)

This test is known as Nin-hydrin test and given by Amino–acid

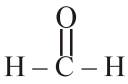
N 

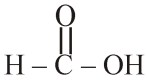
(Yellow–colour)

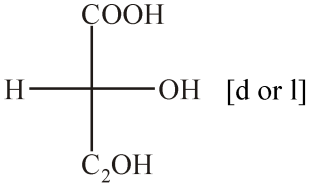
O 

P 

(C.O.S. is present)

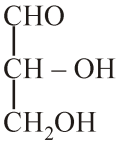
Q 

Q' 

R 
[d or l]

Rankers Academy JEE

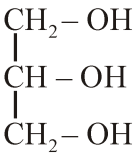
S



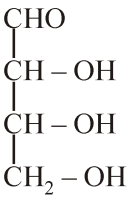
S'



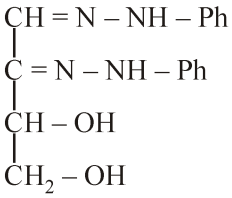
T



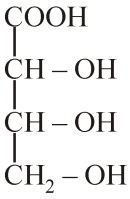
U



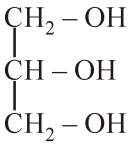
V



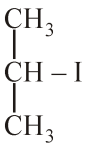
W



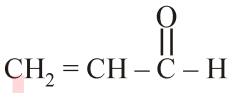
X



Y



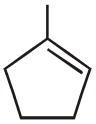
Z

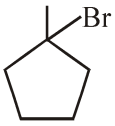


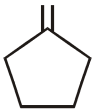
Rankers Academy JEE

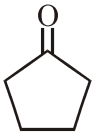
ORGANIC CHEMISTRY

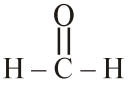
ROADMAP PROBLEM # 14 [SOLUTION]

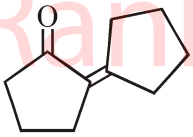
A 

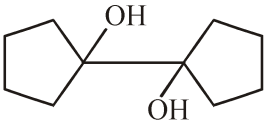
B 

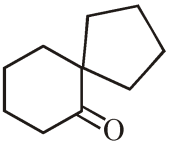
C 

D 

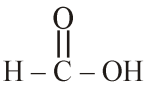
E 

F 

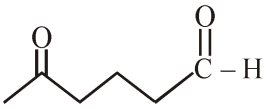
G 

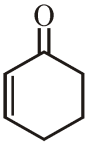
H 

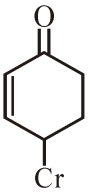
I $CH_3 - OH$

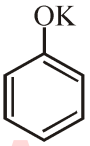
J 

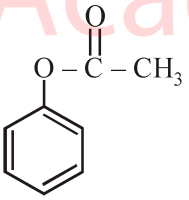
K H_2CO_3

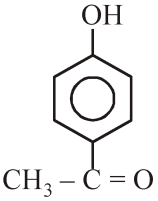
L 

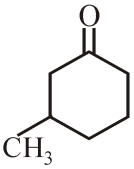
M 

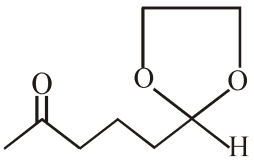
N 

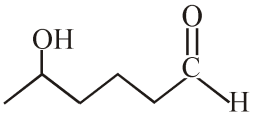
O 

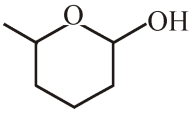
P 

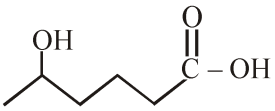
Q 

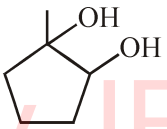
R 

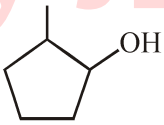
S 

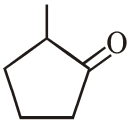
T 

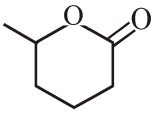
U 

V 

W 

X 

Y 

Z 

Rankers Academy JEE

ANSWER

1. **Ans.B,C**

2. **Ans.B,C,D**

3. **Ans. A → R; B → S; C → P; D → Q**

4. **Ans.** A → T; B → R,S; C → P; D → Q, S

5. **Ans. 3**

6. **Ans. 7**

Sol. N, R, T, U, W, Y, Z

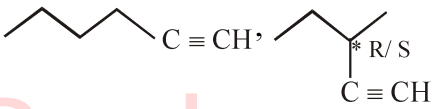
7. **Ans. 8**

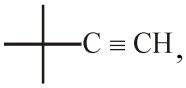
Sol. D, H, M, N, Q, R, T, Y

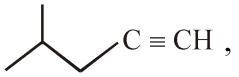
8. **Ans. 5**

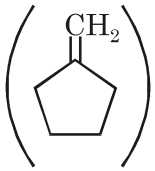
Sol. H, Q, R, T, Y

9. **Ans. 5**







, MF, C_6H_{10},


10. **Ans. 5**

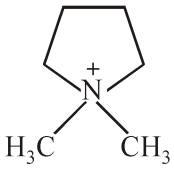
E, J, L, T, U

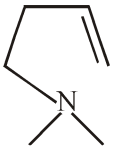
Rankers Academy JEE

ORGANIC CHEMISTRY

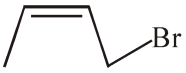
ROADMAP PROBLEM # 15 [SOLUTION]

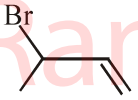
A 

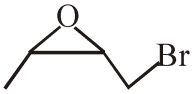
B 

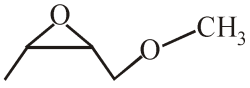
C 

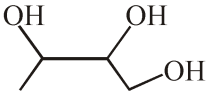
D 

D' 

E 

F 

G 

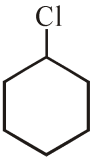
H 

I 

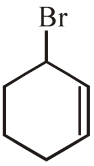
J HCOOH

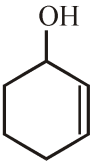
K HCHO

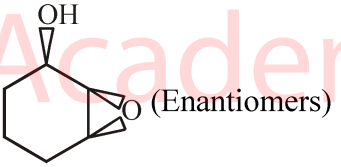
L 

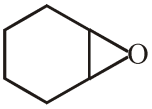
M 

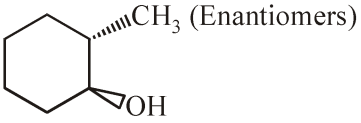
N 

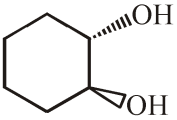
O 

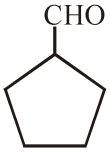
P 

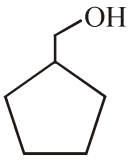
Q 

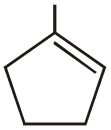
R 

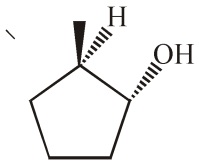
S 

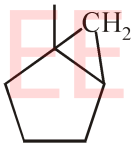
T 

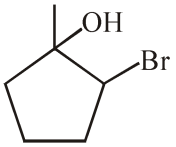
U 

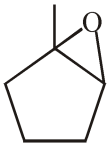
V 

W 

W' 

X 

Y 

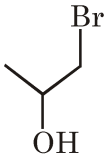
Z 

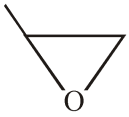
Rankers Academy JEE

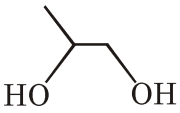
ORGANIC CHEMISTRY

ROADMAP PROBLEM # 16 [SOLUTION]

(A) 

(B) 

(C) 

(D) 

(E) $CH_3 - COONa$

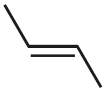
(F) $CHBr_3$

(G) CH_4

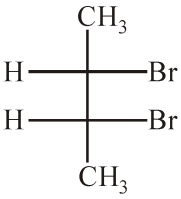
(H) $CH_3 - Cl$

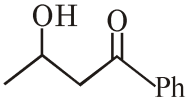
(I) $CH \equiv CH$

(J) $CH_3 - C \equiv C - CH_3$

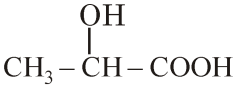
(K) 

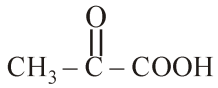
(L) $CH_3 - CHO$

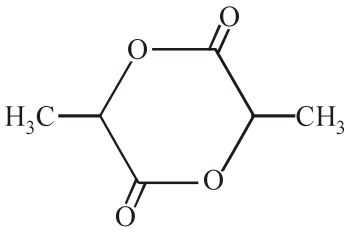
(M) 

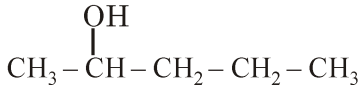
(N) 

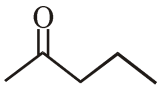
(O) CH_3CH_2OH

(P) 

(Q) 

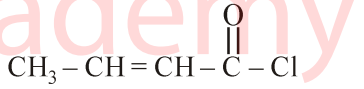
(R) 

(S) 

(T) 

(U) $CH_3 - CH = CH - CHO$

(V) $CH_3 - CH = CH - COOH$

(W) 

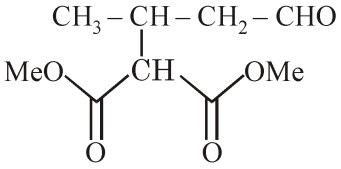
(X) $CH_3 - CH = CH - CHO$

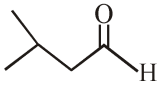
(Y) $CH_3 - CH = CH - CH_2OH$

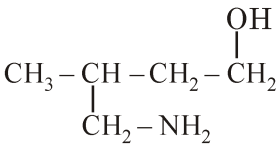
(Z) $CH_3 - CH_2 - CH_2 - CHO$

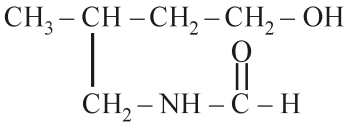
Rankers Academy JEE

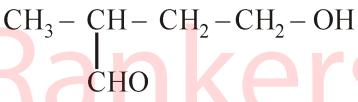
(1) 

(2) 

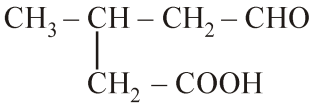
(3) 

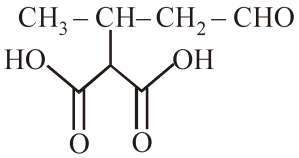
(4) 

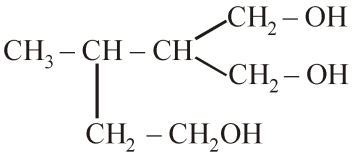
(5) 

(6) 

Rankers Academy JEE

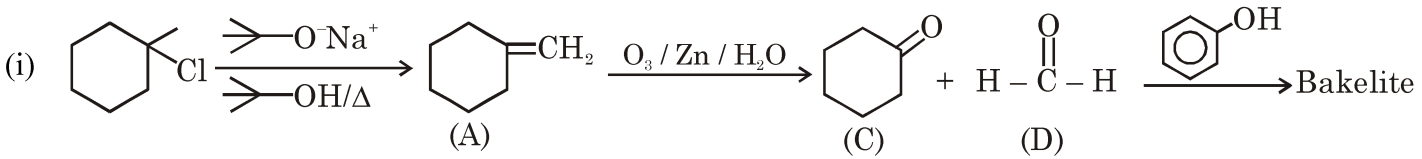
(7) 

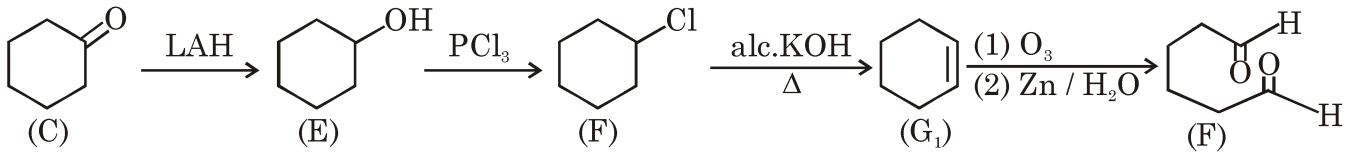
(8) 

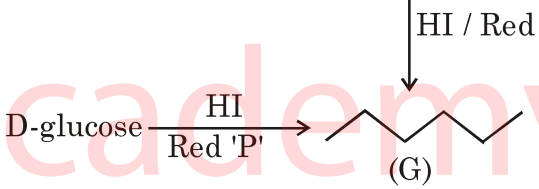
(9) 

ORGANIC CHEMISTRY

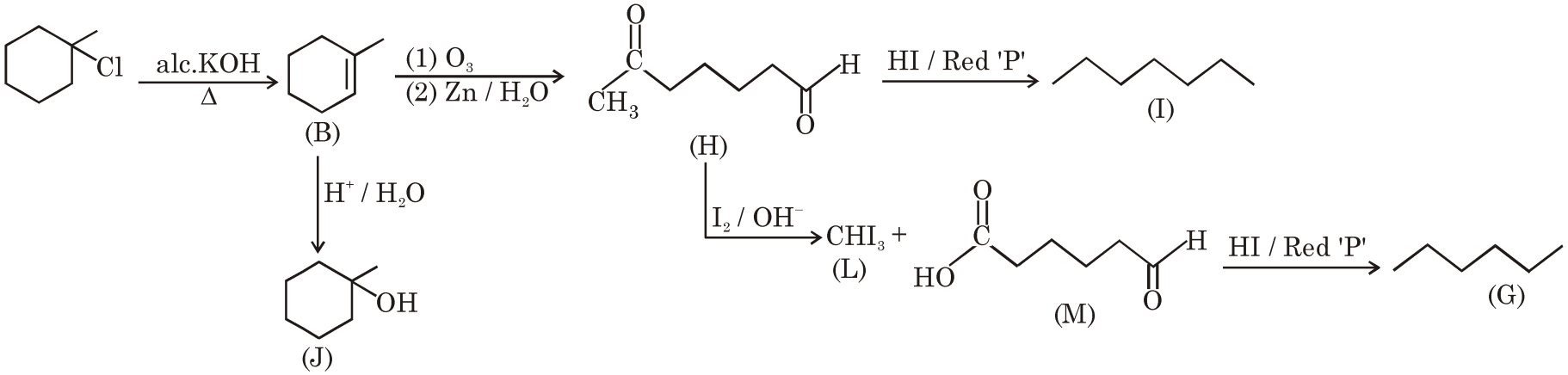
ROADMAP PROBLEM # 17 [SOLUTION]

(i) 





Rankers Academy JEE



(ii) CHI_3 (L) $\xrightarrow[\text{Powder}]{Ag}$ $CH \equiv CH$ (N) $\xrightarrow[Cu]{\text{Red Hot}}$ (O) $\xrightarrow{Cl/AlCl_3}$ (P) $\xrightarrow[(2)\ H^+/H_2O]{(1)\ O_2/h\nu}$ (R) + (Q)

(Q) $\xrightarrow{FeCl_3}$ Violet colour

(Q) $\xrightarrow[KOH]{CHCl_3}$ (S)

(iii) $\xrightarrow{I_2/OH^-}$ CHI_3 (L) + $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-O^-$ (T) $\xrightarrow[\text{Electrolysis}]{NaOH}$ (U) $\xrightarrow{Br_2/h\nu}$ (V) $\xrightarrow[\text{ether}]{Mg}$ (W)

(W) $\xrightarrow[(2)\ H^+/H_2O]{(1)\ H-\overset{O}{\overset{\|}{C}}-H\ (D)}$ (X) $\xrightarrow{PCC}$ (Y) $\xrightarrow[\text{Reagent}]{\text{Tollen's}}$ Ag Mirror + (Z)

Rankers Academy JEE

ORGANIC CHEMISTRY

ROADMAP PROBLEM # 18 [SOLUTION]

(A) $CH_3–CH_2–CH_3$

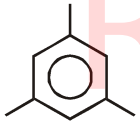
(B) $CH_3–\underset{\underset{Br}{|}}{CH}–CH_3$ (Major)

(C) $\boxed{CH_2=CH–CH_3}$

(D) $CH_2=CH–\underset{\underset{Br}{|}}{CH_2}$

(E) $CH_3–\underset{\underset{Br}{|}}{CH}–\underset{\underset{Br}{|}}{CH_2}$

(F) $CH_3 – C \equiv CH$

(G) 

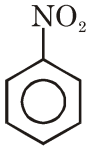
(H) HCHO

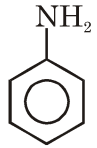
(I) $CH_3 – CHO$

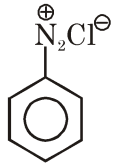
(J) $Ph–CH_2–OH+HCOO^{\ominus}$

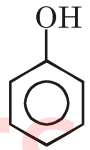
(K) Ph – COOH

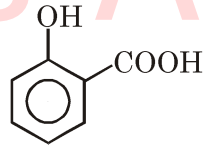
(L) 

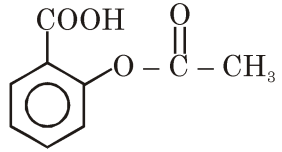
(M) 

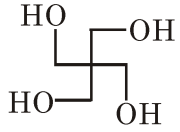
(N) 

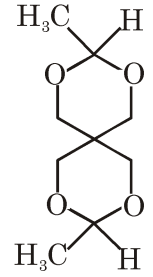
(O) 

(P) 

(Q) 
(Ortho product)

(R) 
(Aspirin)

(S) 

(T) 

(U) $HCO\overset{\ominus}{O}$

(V) $CHBr_3$
(pale yellow)

(W) $H–C\equiv C–H$

(X) $CH_3 – C \equiv C – CH_3$

(Y) 

(Z) 

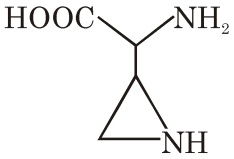
Rankers Academy JEE

ORGANIC CHEMISTRY

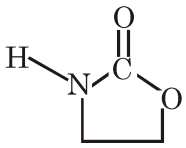
ROADMAP PROBLEM # 19 [SOLUTION]

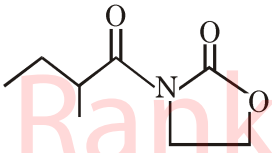
(A) $NH_2 - CH_2 - COOH$

(B) $NH_2 - CH(CH_2 - Ph) - COOH$

(C) 

(D) $NH_2 - CH_2 - CH_2 - OH$

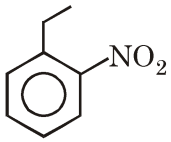
(E) 

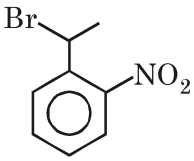
(F) 

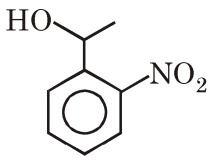
(One – Chiral center)

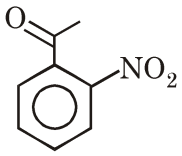
(G) $Ph - CH_2 - CH_2 - NH_2$

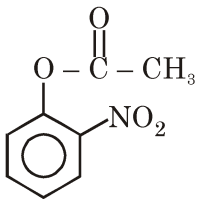
(H) $Ph - CH_2 - CH_3$

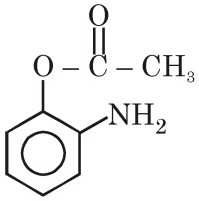
(I) 

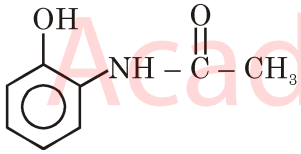
(J) 

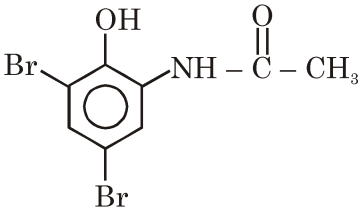
(K) 

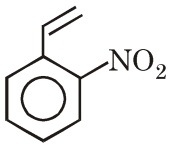
(L) 

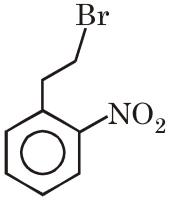
(M) 

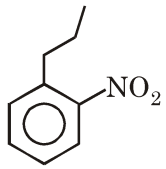
(N) 

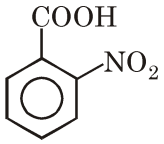
(O) 

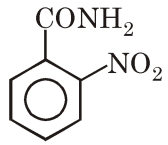
(P) 

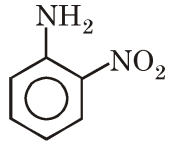
(Q) 

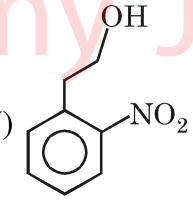
(R) 

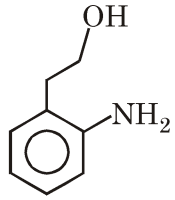
(S) 

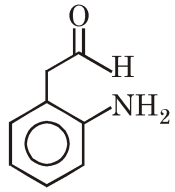
(T) 

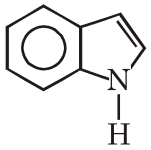
(U) 

(V) 

(W) 

(X) 

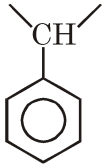
(Y) 

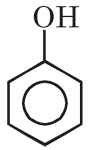
(Z) 

Rankers Academy JEE

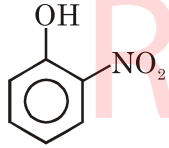
ORGANIC CHEMISTRY

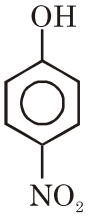
ROADMAP PROBLEM # 20 [SOLUTION]

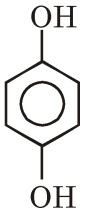
(A) 

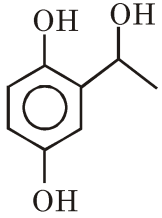
(B) 

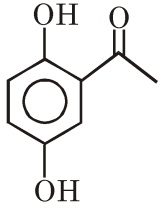
(C) 

(D) 

(E) 

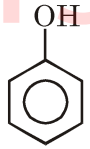
(F) 

(G) 

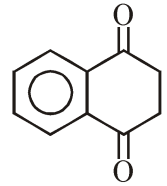
(H) 

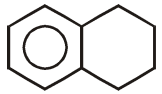
(I) 3

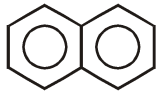
(J) CH_4

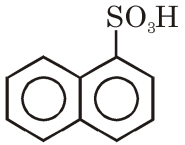
(K) 

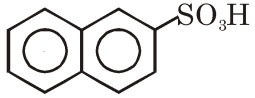
(L) 

(M) 

(N) 

(O) 

(P) 

(Q) 

(R) CHI_3

(S) $CH_3COO^-Na^+$

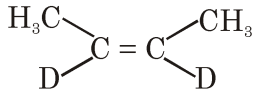
(T) CH_3–Br

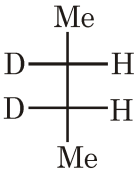
(U) CH ≡ CH

(V) $\underbrace{Ag^+\bar{C}\equiv C^-Ag^+}_{\text{White P.P.T.}}$

(W) $CH_3 - O - CH_3$

(X) $CH_3 - C \equiv C - CH_3$

(Y) 

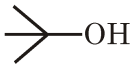
(Z) 

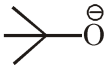
Rankers Academy JEE

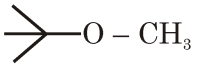
ORGANIC CHEMISTRY

ROADMAP PROBLEM # 21 [SOLUTION]

Rankers Academy JEE

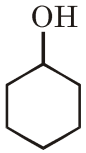
(A_1) 

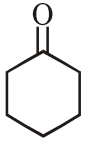
(L_1) 

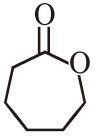
(L_2) 

(E) 

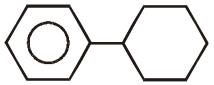
(N) CH_3OH

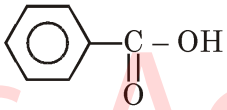
(H_1) 

(A_2) 

(I_1) 

(T) 

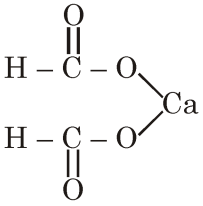
(H_2) 

(O) 

(M_1) $CH_3 - \underset{\underset{O}{\|}}{C} - OH$

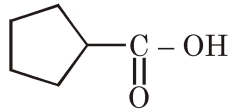
(U) $CH_3 - CH_2 - OH$

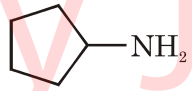
(M_2) $CHCL_3$

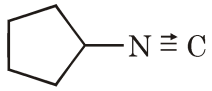
(K) 

(I_2) $H - \underset{\underset{O}{\|}}{C} - H$

(N) $CH_3 - OH$

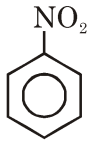
(H_3) 

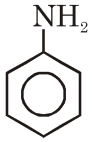
(A_3) 

(I_3) 

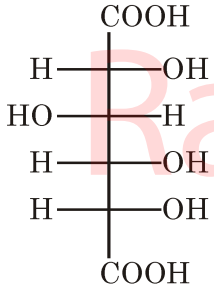
ORGANIC CHEMISTRY

ROADMAP PROBLEM # 22 [SOLUTION]

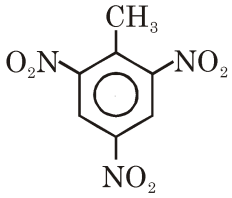
(A) 
(Nitro Benzene)

(B) 
(Anilene)

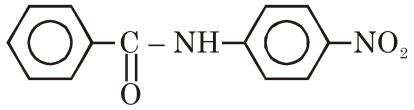
(C) No reaction

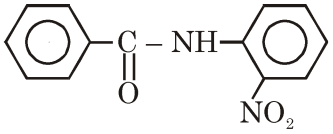
(D) 
(Saccharic acid)

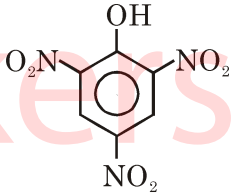
(E) 
n–Hexane

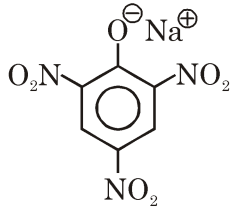
(F) 
(TNT)
(Explosive)

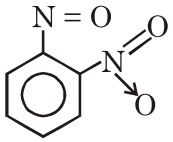
(G) 
(Nitronium ion)

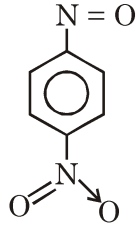
(H) 

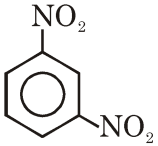
(I) 

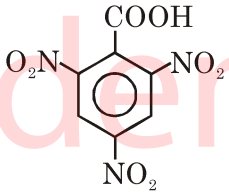
(J) 
2,4,6–Trinitrophenol
(Picric acid)

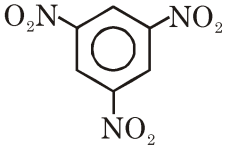
(K) 

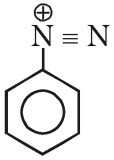
(L) 

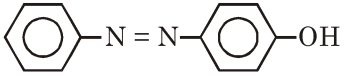
(M) 

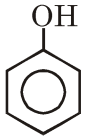
(N) 

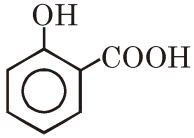
(O) 

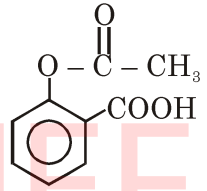
(P) 

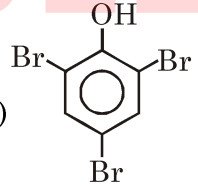
(Q) 

(R) 
p-Hydroxy azo benzene (Azo dye)

(S) 

(T) 

(U) 

(V) 

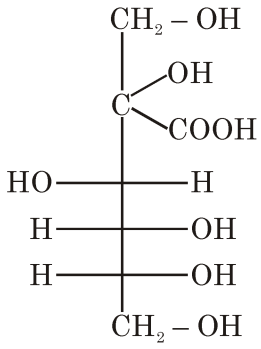
Rankers Academy JEE

ORGANIC CHEMISTRY

ROADMAP PROBLEM # 23 [SOLUTION]

A : Fructose

B :



C : n-Hexane

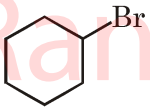
D :



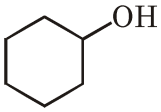
E :



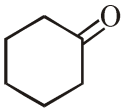
F :



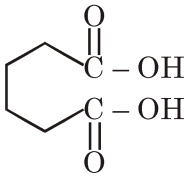
G :



H :



I :



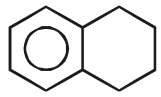
J : $NH_2 - \overset{\overset{O}{\|}}{C} - (CH_2)_4 - \overset{\overset{O}{\|}}{C} - NH_2$

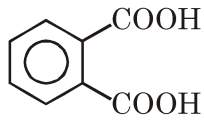
K : $NC - (CH_2)_4 - CN$

L : $NH_2 - CH_2 - (CH_2)_4 - CH_2 - NH_2$

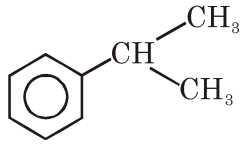
M : Nylon–6,6

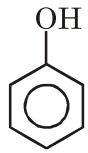
Rankers Academy JEE

N : 

O : 

P : Glyptal (It is not a thermosetting polymer)

Q : 

R : 

S : $CH_3 - \overset{\overset{O}{\|}}{C} - CH_3$

T : Benzaldehyde

U : Toluene

V : CHI_3

W : $CH_3 - COO^-Na^+$

X : $CH_3 - CH_3$

Y : $CH_3 - CH_2 - Cl$

Z : 

A' : $CH_3 - CH_2 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3$

B' : $CH_3 - CH_2 - CH = CH_2$

C' : $CH_3 - CH_2 - CH_2 - CH_2 - OH$

D' : $CH_3 - CH_2 - CH_2 - COOH$

E' : $CH_3 - CH_2 - \underset{\displaystyle Cl}{\underset{|}{CH}} - COOH$

F' : $CH_3 - CH = CH - COO^-$

G' : $CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_2 - COOH$

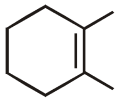
H' : $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_2 - COOH$

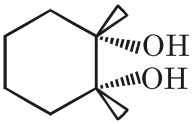
I' : PHBV (Bio-degradable polymer)

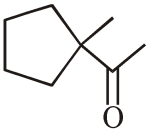
Rankers Academy JEE

ORGANIC CHEMISTRY

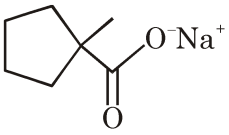
ROADMAP PROBLEM # 24 [SOLUTION]

A : 

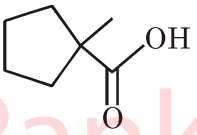
B : 

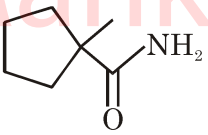
C : 

D : CHI_3

E : 

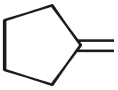
F : $CH \equiv CH$

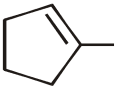
G : 

H : 

I : 

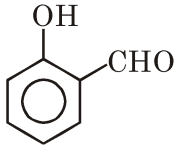
J : 

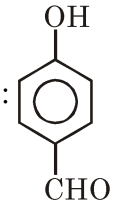
K : 

L : 

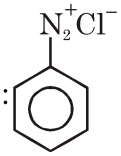
M : 

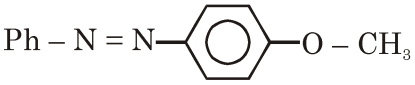
N : Phenol

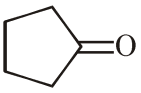
O : 

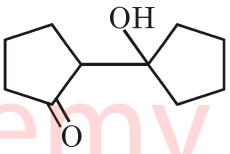
P : 

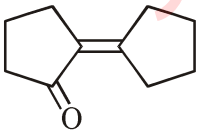
Q : Aniline

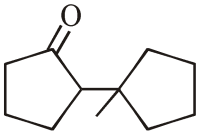
R : 

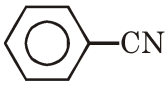
S : 

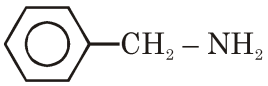
T : 

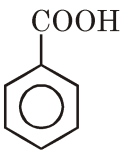
U : 

V : 

W : 

X : 

Y : 

Z : 

Rankers Academy JEE

ROADMAP PROBLEM : 24 BASED QUESTION WITH ANSWER

1. **Ans.** (I), (Q), (Y)

(A) M (B) N (C) Q (D) X

2. **Ans. (M)**

3. **Ans.(A,B)**

4. **Ans.(D)**

5. **Ans.(A,B,C)**

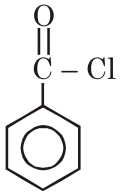
6. Ans. (2)

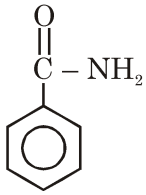
E_{1CB}

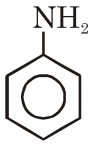
Rankers Academy JEE

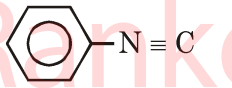
ORGANIC CHEMISTRY

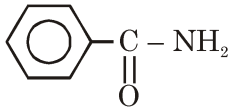
ROADMAP PROBLEM # 25 [SOLUTION]

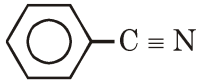
A : 

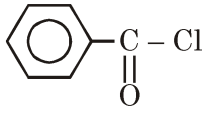
B : 

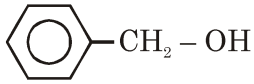
C : 

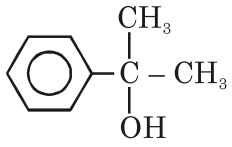
D : 

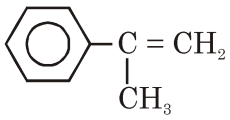
E : 

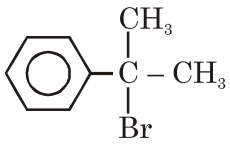
F : 

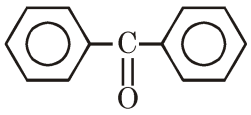
G : 

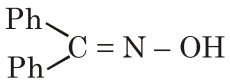
H : 

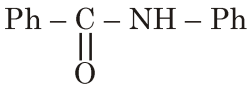
I : 

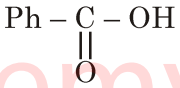
J : 

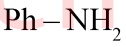
K : 

L : 

M : 

N : 

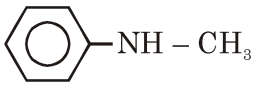
O : 

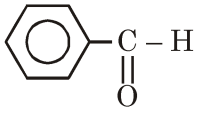
P : 

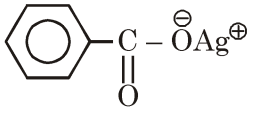
Q : 

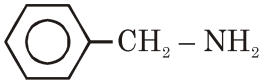
R : 

S : 

T : 

U : 

V : 

W : 

Rankers Academy JEE

ROADMAP PROBLEM : 25 ANSWER

1. **Ans. (3)**

Sol. L, Q, U

2. **Ans. (4)**

Sol. I, IV, V andVI

3. **Ans. (4)**

Sol. 4 moles of KOH is consumed

4. **Ans.(B)**

5. **Ans. (A,B,D)**

6. **Ans. A $\rightarrow$ R, B $\rightarrow$ S, C $\rightarrow$ Q, D $\rightarrow$ P**

7. **Ans. (A,B,C,D)**

8. **Ans.(C)**

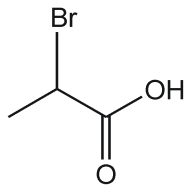
9. **Ans. (D)**

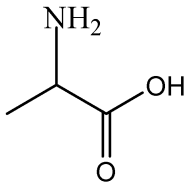
10. **Ans.(A)**

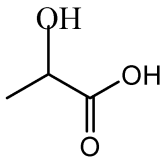
Rankers Academy JEE

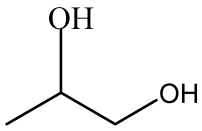
ORGANIC CHEMISTRY

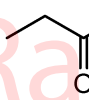
ROADMAP PROBLEM # 26 [SOLUTION]

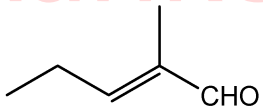
A : 

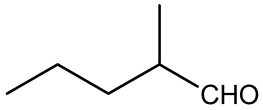
B : 

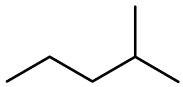
C : 

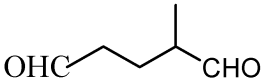
D : 

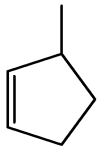
E : 

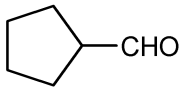
F : 

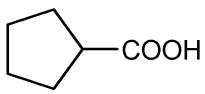
G : 

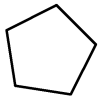
H : 

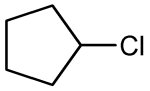
I : 

J : 

K : 

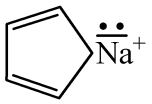
L : 

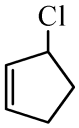
M : 

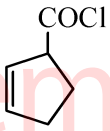
N : 

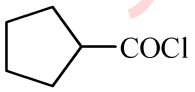
O : 

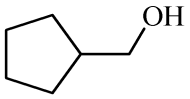
P : 

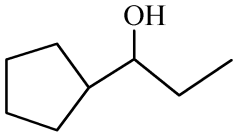
Q : 

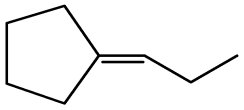
R : 

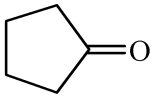
S : 

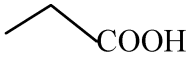
T : 

U : 

V : 

W : 

X : 

Y : 

Z : 

Rankers Academy JEE

ORGANIC CHEMISTRY

ROADMAP PROBLEM # 27 [SOLUTION]

A : 

B : $CH_3-\underset{\underset{CH_3}{|}}{CH}-Cl$

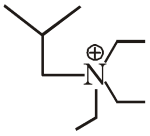
C : $CH_3-\underset{\underset{F}{|}}{CH}-CH_3$

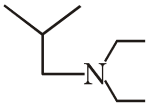
D : $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

E : $CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$

F : $CH_3-\underset{\underset{CN}{|}}{CH}-CH_3$

G : 

H : 

I : 

J : $CH_2=CH_2$

K : 

L : $CH_3-\underset{\underset{COOH}{|}}{CH}-CH_3$

M : $CH_3-\underset{\underset{COOH}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3$

N : $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{CH}}-I$

O : $CH_2=CH-CH=O$

P : $CH_2=CH-CH_2-OH$

Q : $CH_2=CH-CH_2-Cl$

R : $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{Cl}{|}}{CH_2}$

S : $CH_3-C\equiv CH$

T : $CH_3-C\equiv C^{\ominus}Na^{\oplus}$

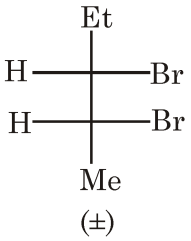
U : $CH_3-C\equiv C-CH_3$

V : $CH_3-CH_2-C\equiv C^{\ominus}Na^{\oplus}$

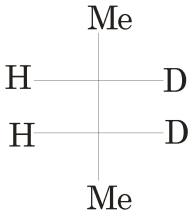
W : $CH_3-CH_2-C\equiv C-CH_3$

Rankers Academy JEE

X : 

Y : 

(±)

Z : 

Meso

ROADMAP PROBLEM : 27 BASED QUESTION WITH ANSWER

1. Ans.(A,B,C,D)
2. Ans.(A,B,C,D)
3. Ans.(B,C,D)
4. Ans.(4)
5. Ans.(1)
6. Ans.(2)
7. Ans. (9)
8. Ans. (A,B,C)
9. Ans. A – S; B–T; C–P; D–Q
10. Ans. A $\rightarrow$ S; B $\rightarrow$ S; C $\rightarrow$ S; D $\rightarrow$ T

Rankers Academy JEE